Modified.

* Suggested Modifications:
1) Count comparisions made during the search.
2) Search for multiple key in one execution.
3) Compare search path lengths in AVL tree and ano ordinary BST.

* Problem statement:
Write a C++ program to construct an AVL tree, search for a specific key, and display the path followed during the search. The program should count the no of comparisions made during searching, support searching from multiple key in one execution, and compare search path length in an AVL tree and ano ordinary Binary search trees. (BST)

* Solution.
An AVL tree is a self Balancing Binary search Tree (BST) that maintains its balance using rotations during insertion. This helps keep the tree height small and makes searching efficient.

* Objectives:
1) constructs an AVL tree and an ordinary BST using the same input elements.
2) searches for a specified key and display the path followed.
3) counts the no of comparisions made during each search.
4) support searching for multiple keys in a single execution.
5) compares the search path length of the AVL tree

and the ordinary BST.

* Algorithm:

1) Start.
2) Create an empty AVL tree and an empty BST.
3) Read the no of elements.
4) Insert each element into both trees.
5) Maintain the AVL tree's Balance using rotations.
6) Read the NO OF keys to search.
7) Read the first search key.
8) Initialize the comparision counter and path length for both trees to zero.
9) Start searching from the root of the AVL tree.
10) Display each visited Node and count each key comparision.
11) Move to the left child if the search key is smaller; otherwise, move to the right child.
12) Repeat until the key is found or a NULL Node is reached.
13) Display the AVL search path, search se result, comparision count, and path length.
14) Repeat the same searching procedure for the ordinary BST.
15) Display the BST search path, search result, comparision count, and path length.
16) Compare the search path lengths of the two trees.
17) If more key remaining, read the next key and repeat steps 8-16.
18) Stop.

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* Flowchart :-

